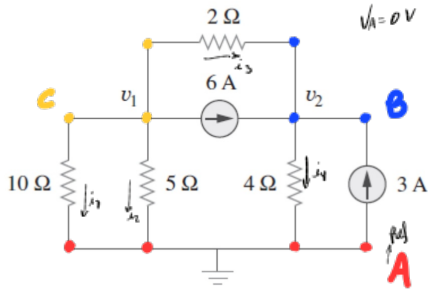


3.2 Para o circuito da Figura 3.51, obtenha  $v_1$  e  $v_2$ .



$$i_1 = \frac{V_C}{10} \quad i_2 = \frac{V_C}{5} \quad i_3 = \frac{V_C - V_B}{2} \quad i_4 = \frac{V_B}{4}$$

$$\text{Nó C: } i_1 + i_2 + i_3 = 0$$

$$\text{Nó B: } i_3 = i_4$$

$$\frac{V_C}{10} + \frac{V_C}{5} + \frac{V_C - V_B}{2} = 0 \quad (\times 10)$$

$$9 + \frac{V_C - V_B}{2} = \frac{V_B}{4} \quad (\times 4)$$

$$V_C + 2V_C + 6 + 5V_C - 5V_B = 0$$

$$36 + 2V_C - 2V_B = V_B$$

$$8V_C - 5V_B = -60$$

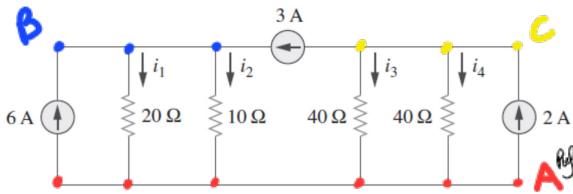
$$2V_C - 3V_B = -36$$

$$V_B = 12V$$

$$V_A = 0V$$

$$V_C = 0$$

3.4 Dado o circuito da Figura 3.53, calcule as correntes  $i_1$  a  $i_4$ .



$$i_1 = \frac{V_B}{20} \quad i_2 = \frac{V_B}{10} \quad i_3 = \frac{V_C}{40} \quad i_4 = \frac{V_C}{40}$$

$$\text{Nó C: } i_3 + i_4 + 3 = 2$$

$$\text{Nó B: } \frac{V_B}{20} + \frac{V_B}{10} = 9 \quad (\times 20)$$

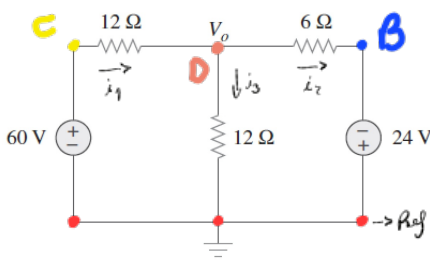
$$2V_C = -40$$

$$3V_B = 180$$

$$V_C = -20$$

$$V_B = 60V$$

3.11 Determine  $V_o$  e a potência dissipada em todos os resistores no circuito da Figura 3.60.



$$i_1 = \frac{V_C - V_D}{12} \quad i_2 = \frac{V_D - V_B}{6} \quad i_3 = \frac{V_D}{12}$$

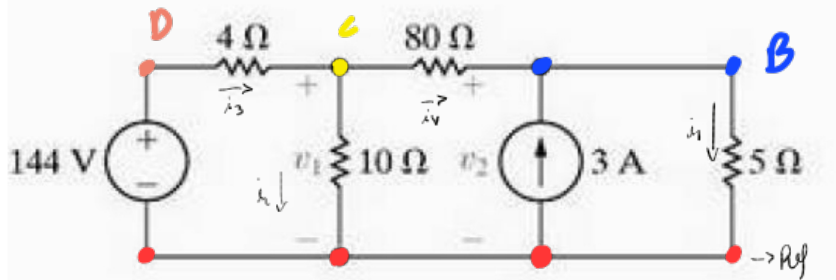
$$V_B = -24V \quad V_C = 60V$$

$$V_o = V_D = 3V$$

$$\text{Nó D: } i_1 = i_2 + i_3$$

$$\frac{60 - V_D}{12} = \frac{V_D - 24}{6} + \frac{V_D}{12} \quad (\times 12) \rightarrow 60 - V_D = 2V_D + 48 - V_D \rightarrow 4V_D = 12 \rightarrow V_D = 3$$

4.12) Use o método das tensões de nó para determinar  $v_1$  e  $v_2$  no circuito a seguir:



$$V_D = 144V$$

$$i_1 = \frac{V_B}{5} \quad i_2 = \frac{V_C}{10} \quad i_3 = \frac{144 - V_C}{4} \quad i_4 = \frac{V_C - V_B}{80}$$

$$\text{Nó B: } \frac{V_C - V_B}{80} + 3 = \frac{V_B}{5} \quad (\times 80)$$

$$\text{Nó C: } i_3 = i_4 + i_1$$

$$\frac{144 - V_C}{4} = \frac{V_C - V_B}{80} + \frac{V_C}{10} \quad (\times 80)$$

$$V_C - V_B + 240 = 16V_B$$

$$V_C - 17V_B = -240$$

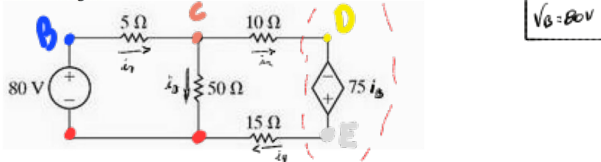
$$2880 - 20V_C = V_C - V_B + 8V_C$$

$$2880 = 29V_C - V_B$$

$$V_C = 100V$$

$$V_B = 20V$$

4.19) Use o método das tensões de nó para calcular a potência gerada pela fonte de tensão dependente no circuito a seguir:



$$V_o = 80V$$

$$i_1 = \frac{80 - V_C}{5} \quad i_2 = \frac{V_C - V_D}{10} \quad i_3 = \frac{V_C}{50} \quad i_4 = \frac{V_E}{15}$$

$$V_C - V_D = 75 \frac{V_C}{50} \quad \text{Suponha: } i_1 = i_4 \rightarrow \frac{V_C - V_D}{5} = \frac{V_E}{15}$$

$$V_E - V_D = \frac{2}{3} V_C$$

$$2V_E - 2V_D - 3V_C = 0$$

$$V_C = 50 \quad V_D = 0 \quad V_E = 15$$

$$\text{Nó C: } i_1 = i_2 + i_3 \rightarrow \frac{80 - V_C}{5} = \frac{V_C - V_D}{10} + \frac{V_C}{50} \quad (\times 50)$$

$$800 - 20V_C = 6V_C - 5V_D$$

$$-16V_C + 5V_D = -800$$

B.

 $i_1$  $i_3$ 

C

$$i_1 = \frac{V_B}{6}$$

$$i_2 = \frac{V_B - V_C}{10}$$

$$i_3 = \frac{V_C}{8}$$

$$i_4 = \frac{V_C}{3}$$

$$* \text{Nó B: } 0 = i_1 + i_2$$

$$\frac{V_B}{6} + \frac{V_B - V_C}{10} = 0 \quad (\times 30)$$

$$5V_B + 3V_B - 3V_C = 0$$

$$8V_B - 3V_C = 0$$

$$* \text{Nó C: } 4 + i_2 + i_3 + i_4$$

$$4 + \frac{V_B - V_C}{10} = \frac{V_C}{8} + \frac{V_C}{3} \quad (\times 120)$$

$$480 + 12V_B - 12V_C = 15V_C + 40V_C$$

$$67V_C - 12V_B = 480$$

$$V_C = 27V$$

$$V_B = 72V$$

4.25) Use o método das tensões de nó para determinar o valor de  $v_o$  no circuito a seguir:

